AI COMMUNICATION SYSTEMS STUDY KIT AUTOMATION GUIDE

A beginner-friendly guide to automating AM/FM simulation, graph generation, and study-kit outputs using Python.

Student: _____
Department: Electronics and Communication Engineering (ECE)
College: _____

Note: Fill in your own details and include only features you actually run and verify.

1. What is Automation?

Automation means using a program or tool to perform repeated steps automatically, with less manual work. In this project, Python can generate the message signal, carrier signal, AM signal, FM signal, and their plots in one run.

2. Project Objective

To create a simple study kit that helps students learn communication-system basics through explanations, simulated waveforms, comparison notes, and quiz questions. The automation part runs the simulation and saves the output graphs to files.

3. Automated Workflow

Step	Action	Output
1	Set simulation parameters	Message/carrier frequencies and modulation settings
2	Generate message and carrier	Reference waveforms
3	Calculate AM and FM	Modulated signals
4	Plot each waveform	AM/FM graph figures
5	Save figures automatically	PNG files in the output folder
6	Review and add to report	Project evidence for submission

4. Python Automation Example

This example shows how to save four graphs in one execution. Run it in Google Colab, Jupyter, or another Python environment that has NumPy and Matplotlib installed. If your Colab usage limit is reached, use a local Python installation or another compatible notebook service.

```
import numpy as np
import matplotlib.pyplot as plt
from pathlib import Path

# Create a folder for saved graph images
output = Path("communication_graphs")
output.mkdir(exist_ok=True)

# Simulation settings
fs = 100_000 # samples per second
t = np.arange(0, 0.003, 1/fs)
fm = 1_000 # message frequency (Hz)
fc = 10_000 # carrier frequency (Hz)
Ac = 1.0 # carrier amplitude
mu = 0.7 # AM modulation index (0 to 1)
beta = 5.0 # FM modulation index

message = np.cos(2*np.pi*fm*t)
carrier = Ac*np.cos(2*np.pi*fc*t)
am = Ac*(1 + mu*message)*np.cos(2*np.pi*fc*t)
fm_signal = Ac*np.cos(2*np.pi*fc*t + beta*np.sin(2*np.pi*fm*t))

signals = [
("Message Signal", message, "message.png"),
("Carrier Signal", carrier, "carrier.png"),
("AM Signal", am, "am_signal.png"),
("FM Signal", fm_signal, "fm_signal.png"),
]
```

```
for title, signal, filename in signals:
plt.figure(figsize=(9, 3))
plt.plot(t*1000, signal)
plt.title(title)
plt.xlabel("Time (ms)")
plt.ylabel("Amplitude")
plt.grid(True)
plt.tight_layout()
plt.savefig(output / filename, dpi=150)
plt.close()

print("Done! Graphs saved in:", output.resolve())
```

Expected output files: message.png, carrier.png, am_signal.png, and fm_signal.png inside the communication_graphs folder.

5. What the Graphs Should Show

Message signal: The low-frequency information waveform.

Carrier signal: A higher-frequency sinusoidal waveform used to carry information.

AM signal: The carrier amplitude follows the message waveform; the carrier frequency remains the same.

FM signal: The carrier's instantaneous frequency varies with the message; its ideal amplitude remains constant.

6. Automation vs. AI

Automation follows programmed steps, such as calculating signals and saving graphs. AI usually refers to systems that learn patterns, generate responses, or make predictions. A Python script that generates AM/FM graphs is automation; it should not be described as an AI model unless an actual AI feature is implemented.

7. Testing Checklist

■ Run the code without errors.

■ Confirm that all four PNG graph files are created.

■ Open the AM graph and check that its envelope follows the message signal.

■ Open the FM graph and check that wave spacing changes while amplitude stays approximately constant.

■ Add your actual graphs and screenshots to the final report or presentation.

■ Include a GitHub repository link only after uploading the files and checking that the link opens.

8. Viva Questions and Short Answers

1. What is automation? Using software to perform repeated tasks with less manual effort.

2. What is modulation? Changing a carrier signal property according to the information signal.

3. What changes in AM? Carrier amplitude changes according to the message.

4. What changes in FM? Instantaneous carrier frequency changes according to the message.

5. What is NumPy used for? Numerical calculations and creating arrays of sample values.

6. What is Matplotlib used for? Drawing and saving signal graphs.

7. Why save graphs as PNG? They can be added to reports, slides, and documentation.

8. Is graph generation alone an AI model? No. It is programmed simulation/automation unless an actual AI feature is added.

9. Suggested Submission Folder

```
AI-Communication-Systems-Study-Kit/
■■■ README.md
■■■ simulation.py (or your notebook)
■■■ graphs/
■ ■■■ message.png
■ ■■■ carrier.png
■ ■■■ am_signal.png
■ ■■■ fm_signal.png
■■■ report.pdf
■■■ presentation.pdf
```

Final reminder: Use the actual graphs and code that you generated. Check all files before submission; this guide is a template and does not prove that your project has been executed successfully.